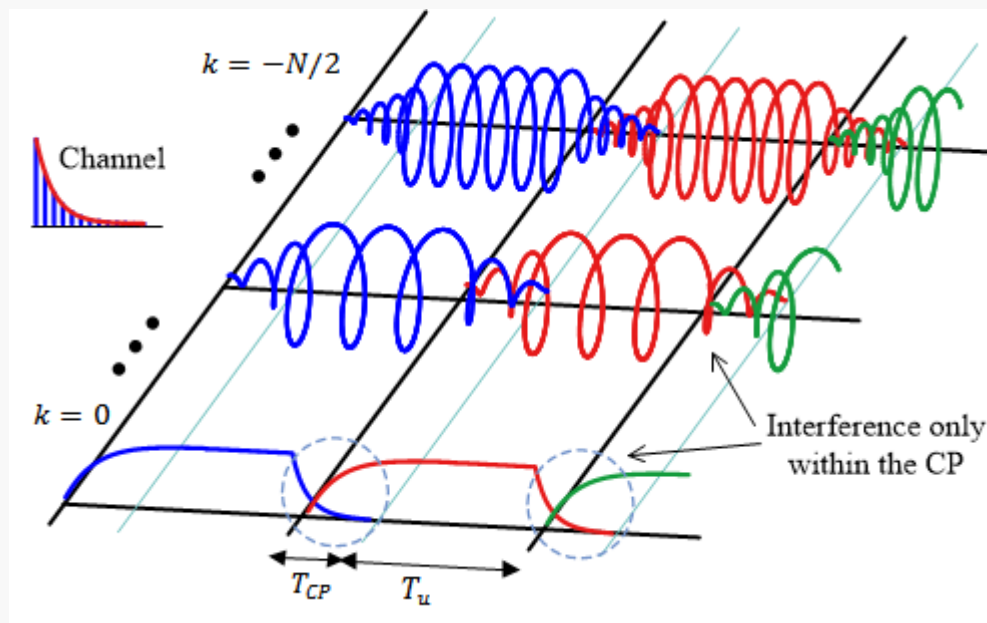


OFDM

Qasim Chaudhari



Introduction

High data rate communication systems

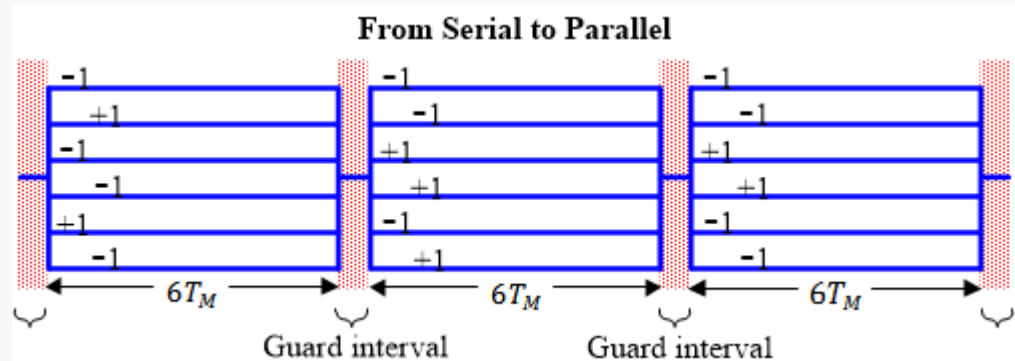
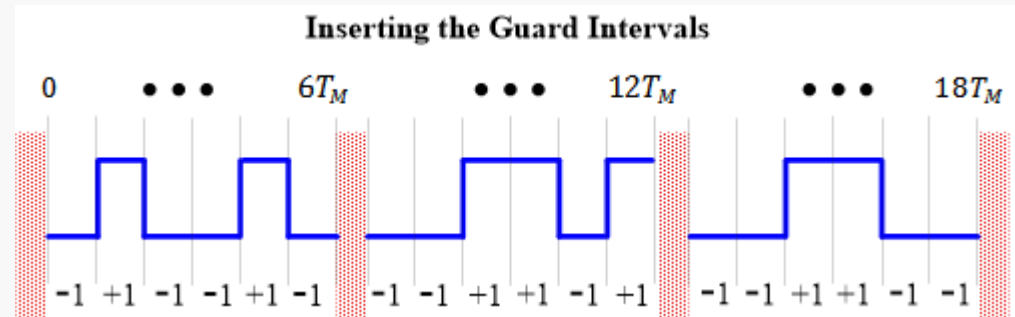
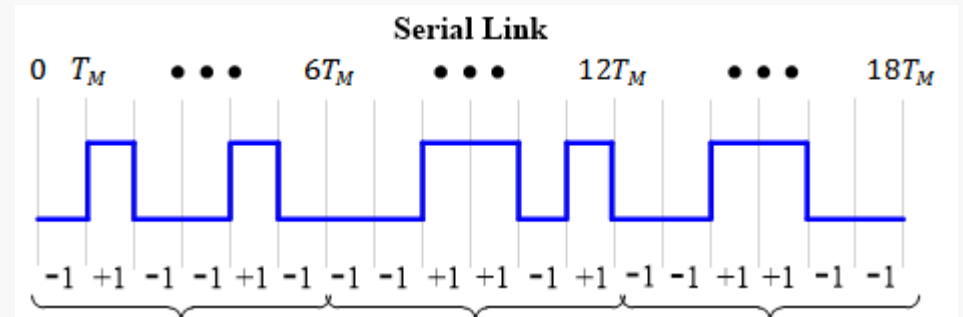
- WiFi (IEEE 802.11a, g, n, ac)
- WiMAX (IEEE 802.16)
- 4G LTE
- 5G



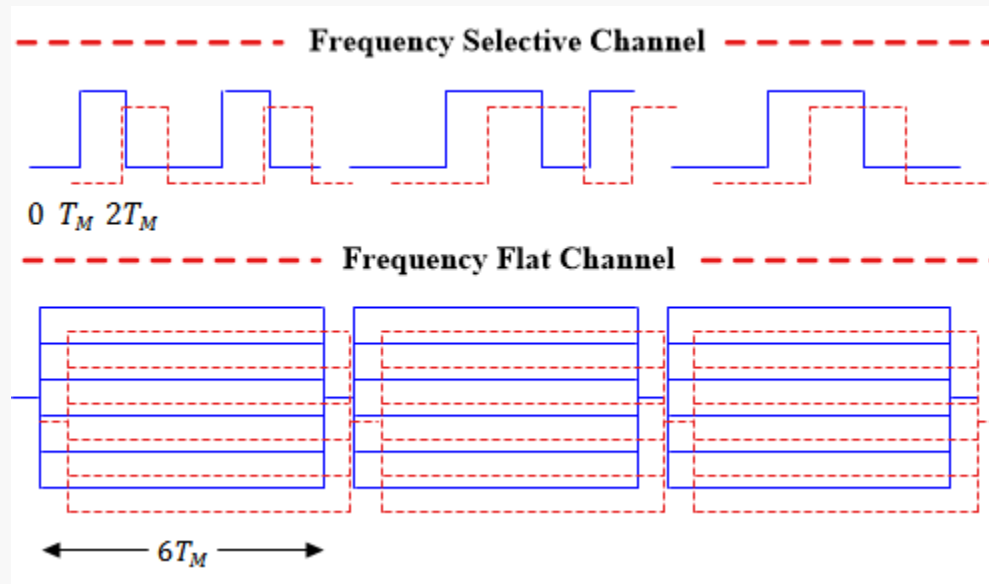
Opposite development to wired communication systems

How OFDM Works: Time

Assume that we have access to a
parallel wires in the air



Home OFDM Works: Time

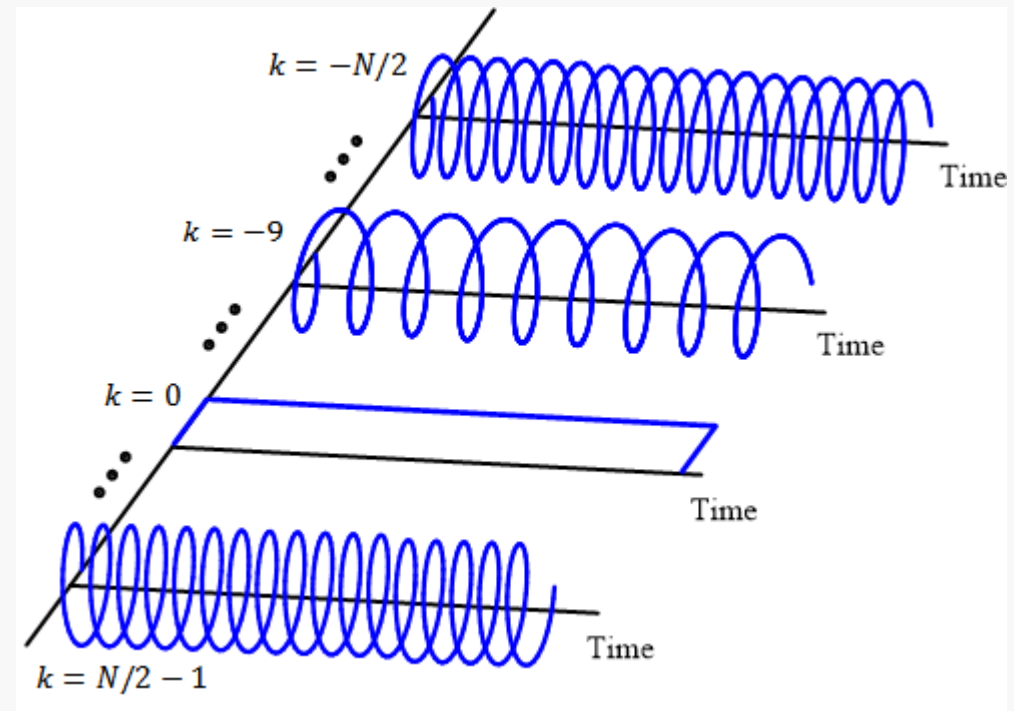


By introducing parallelism, we have fooled a frequency selective channel into behaving as a frequency flat channel!

Home OFDM Works: Time

How to create parallel wires in the air?

- We employ complex sinusoids (called subcarriers)!
- One full cycle in N samples $= 1/NT_S$ Hz



How OFDM Works: Time

Set of complex sinusoids

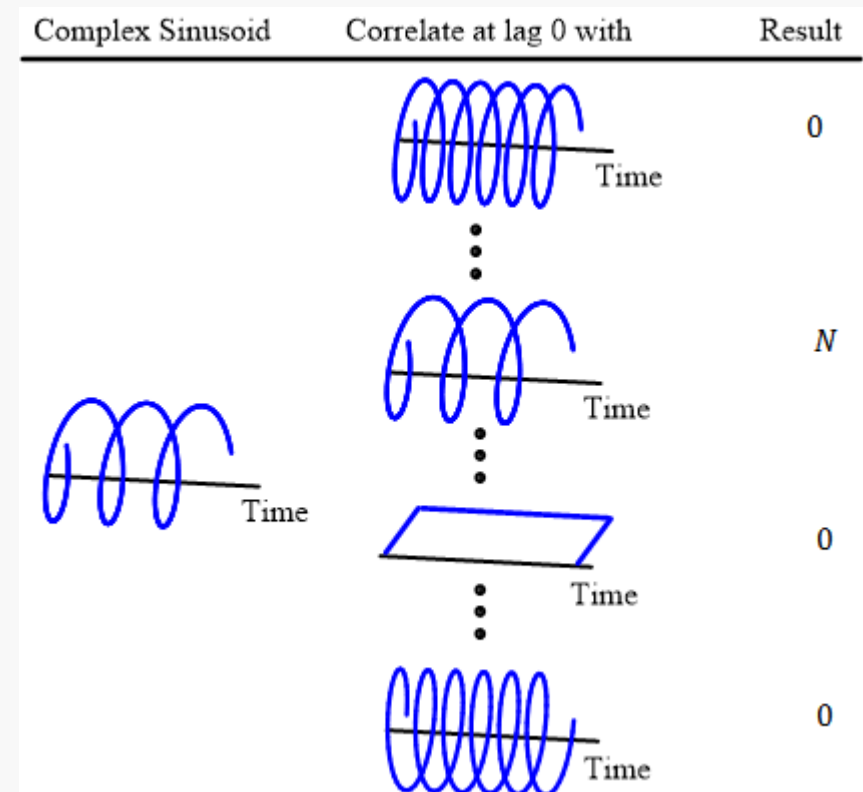
- Orthogonality implies that their correlation at time shift zero is 0

- Let us define $T_u = NT_S$

- So $F_1 = 1/T_u$

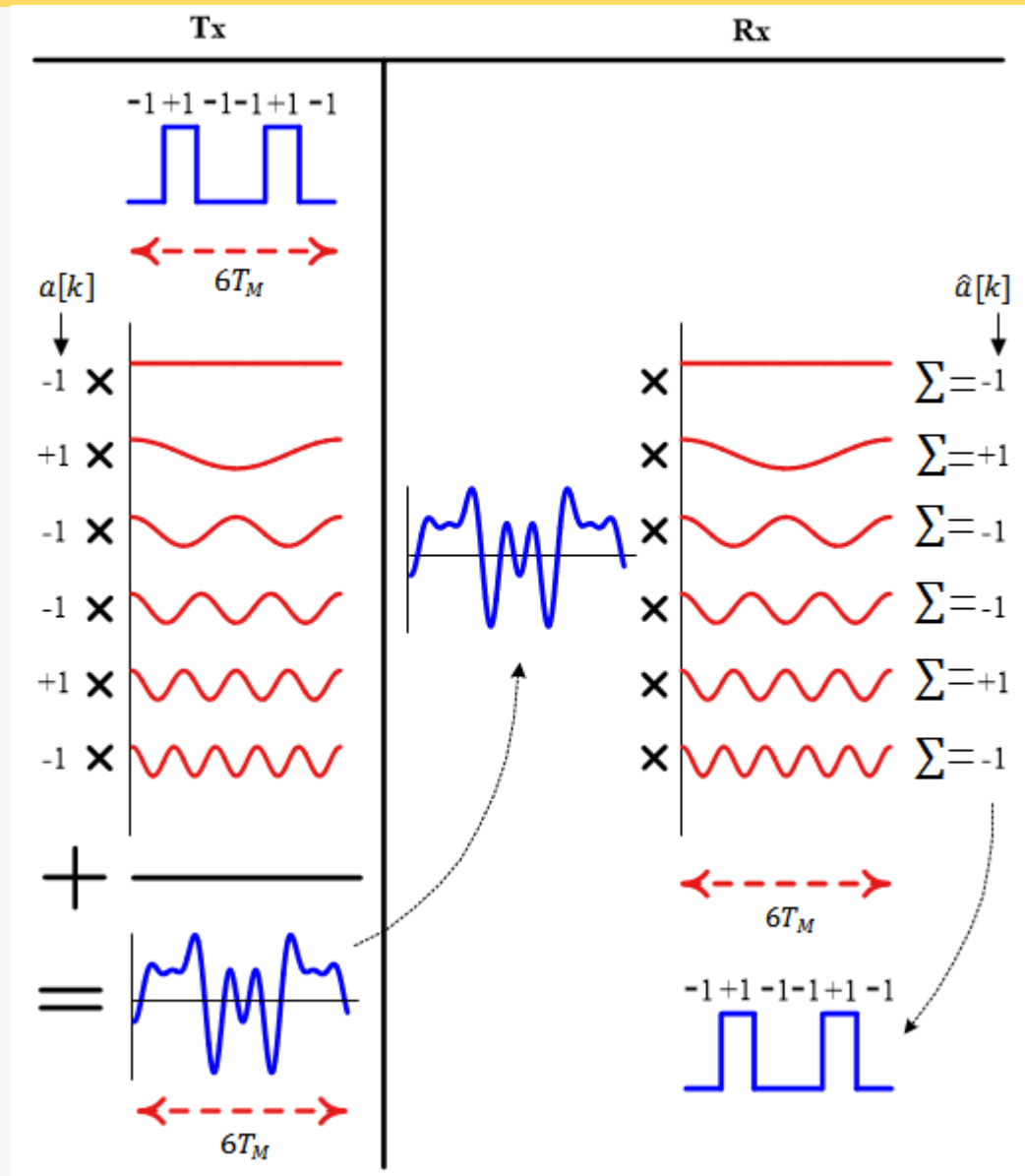
- Remaining frequencies are

$$F_k = kF_1 = \frac{k}{NT_S} = \frac{k}{T_u}$$



Home OFDM Works: Time

What happens when data symbols are modulated?

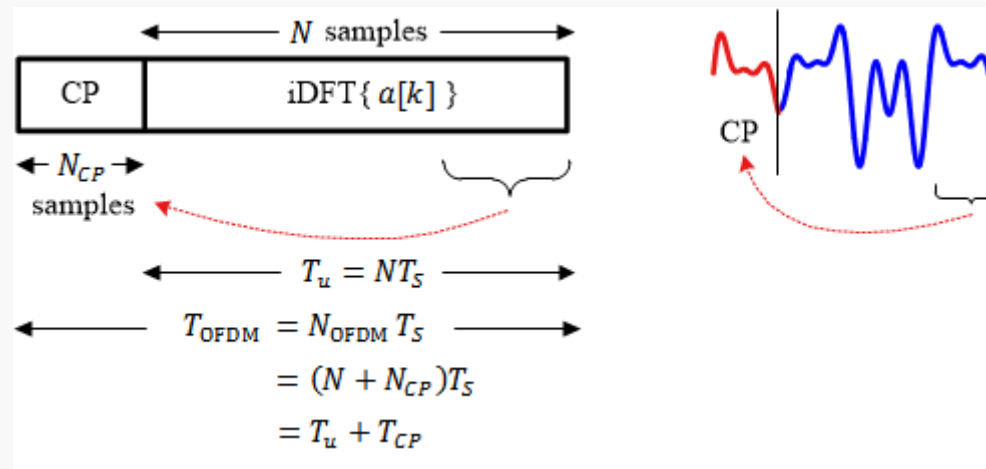


Home OFDM Works: Time

Inserting the Cyclic Prefix (CP)

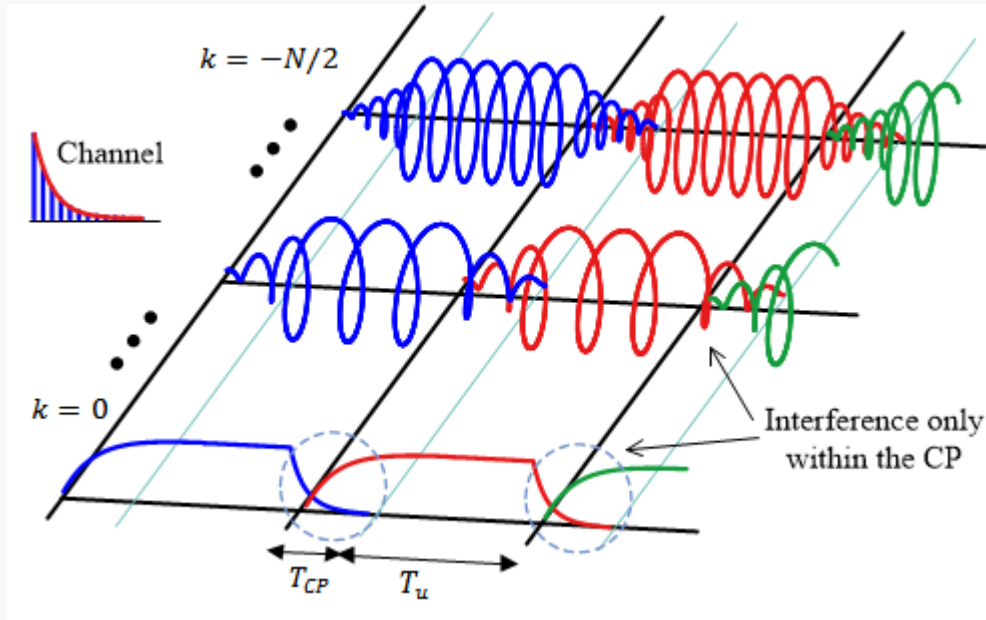
$$N_{\text{OFDM}} = N + N_{\text{CP}}$$

1. Inter-Symbol Interference (ISI)
2. Linear convolution to circular convolution

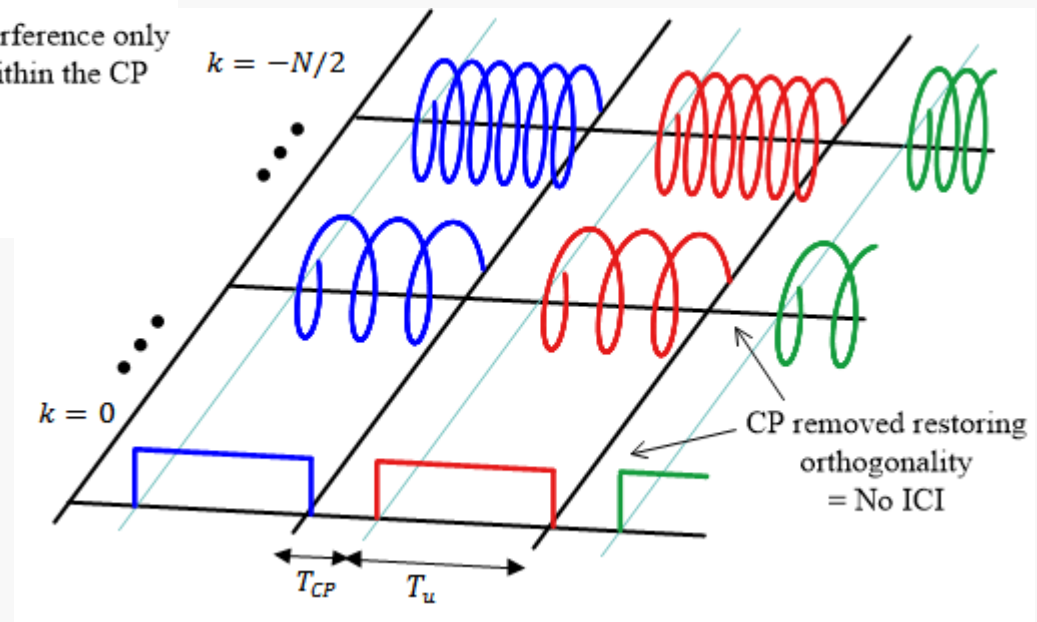


How OFDM Works: Time

3. Inter-Carrier Interference (ICI)

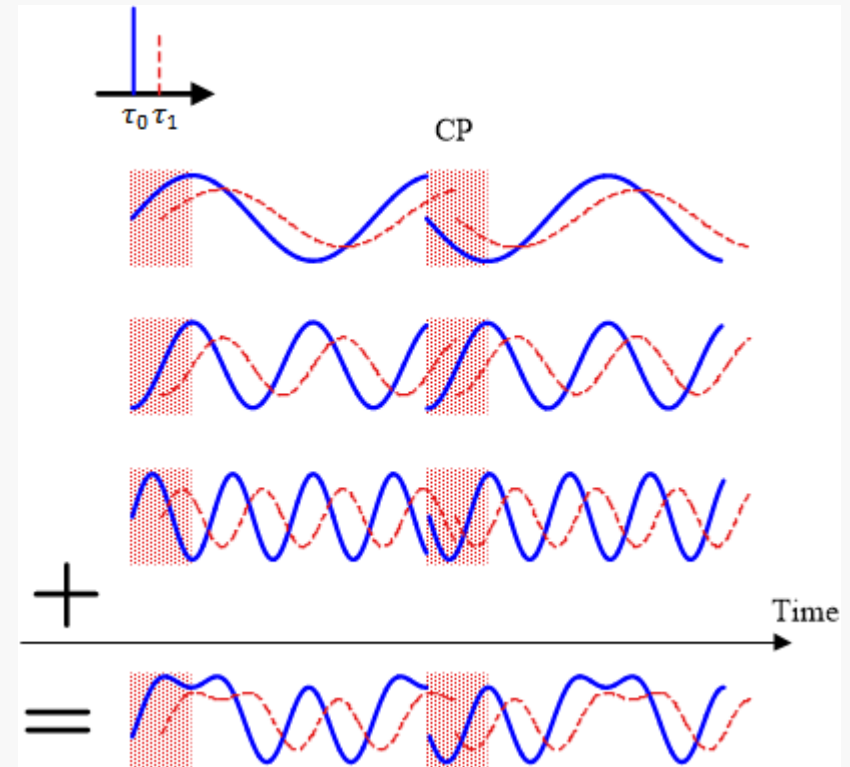


Why equalization becomes simpler



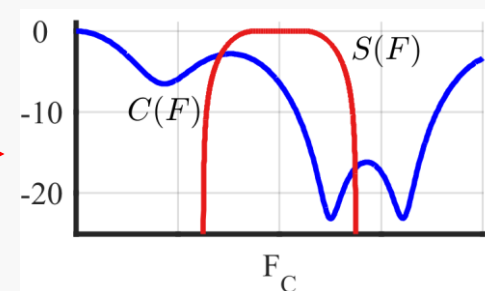
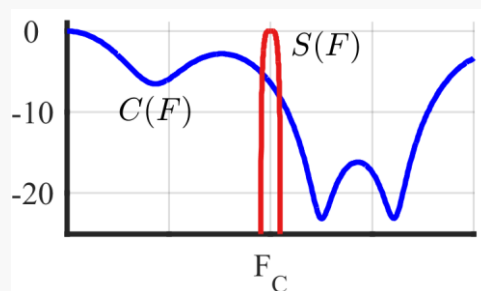
How OFDM Works: Time

Summary: OFDM transforms one high rate serial stream into several low rate parallel streams



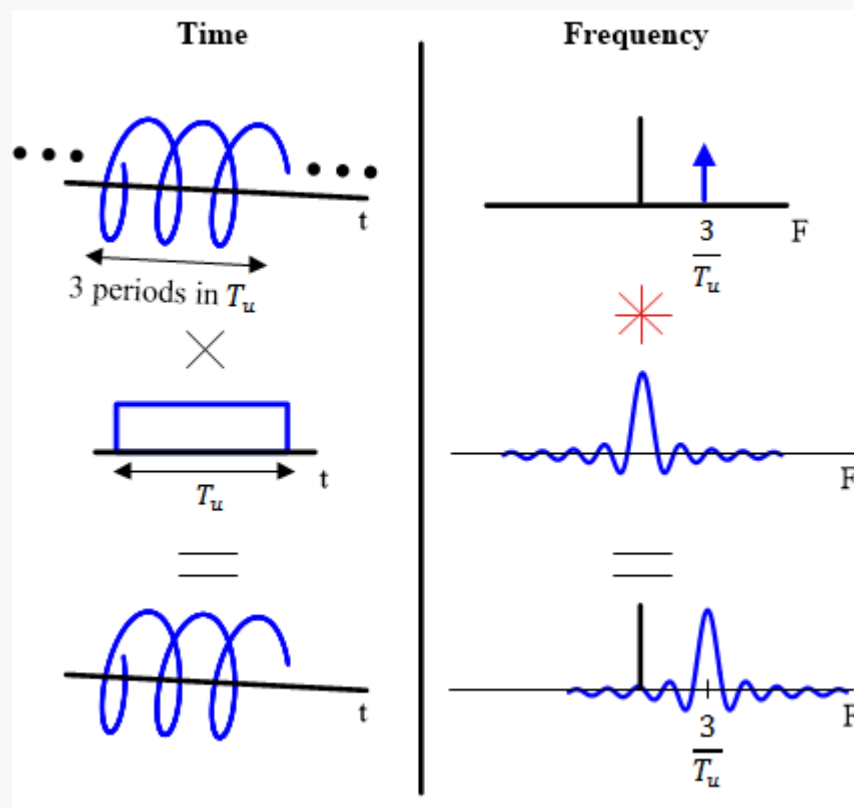
How OFDM Works: Frequency

Increasing the symbol rate



Have to equalize a wider spectrum!

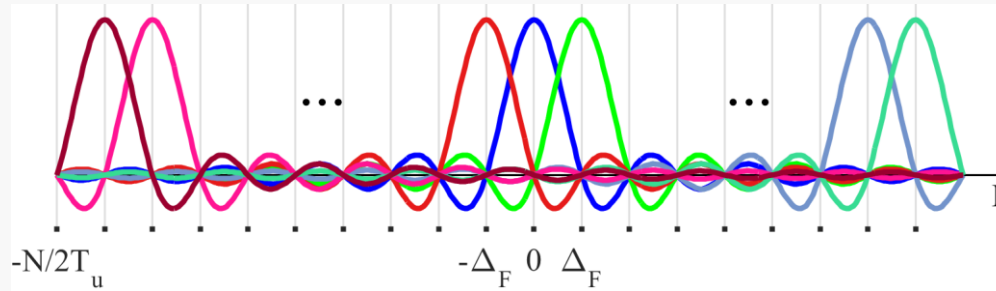
Subcarriers in frequency domain?



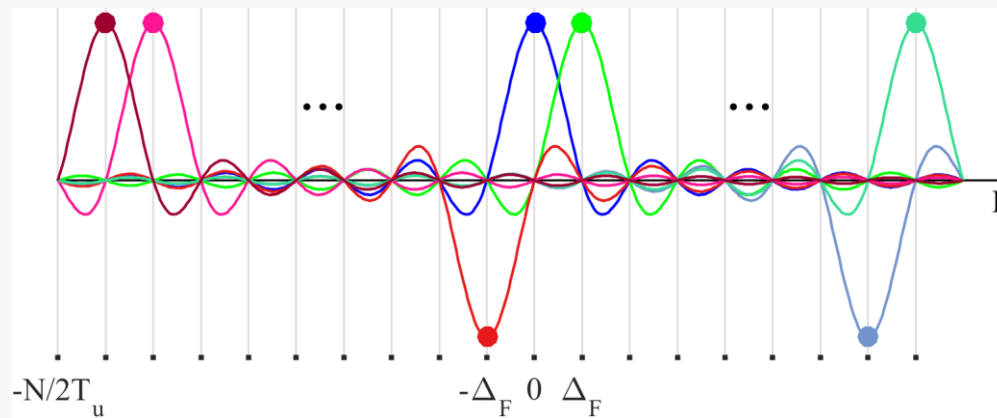
How OFDM Works: Frequency

The complete show

- Unmodulated



- Modulated

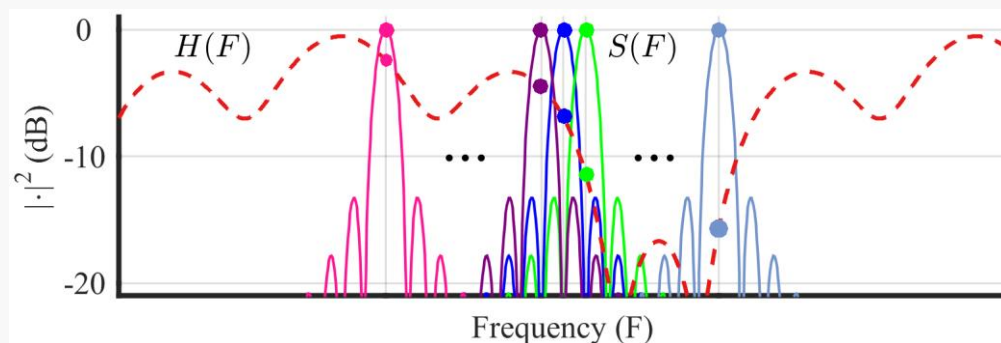
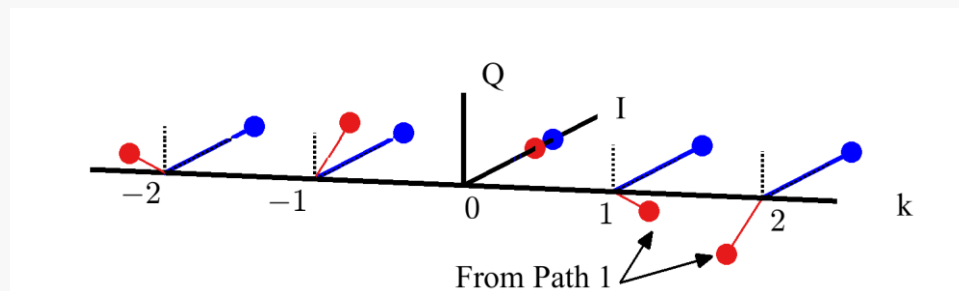


How OFDM Works: Frequency

Why equalization becomes simpler?

- Each subcarrier is affected by its own channel
- Or the subcarriers sample the channel at N locations in frequency domain!
- For each subcarrier k

$$Z[k] = a[k] \cdot H[k]$$



$$\begin{aligned} s(t) &\longrightarrow S(F) \\ s(t - t_0) &\longrightarrow S(F) \cdot \exp(-j2\pi F t_0) \end{aligned}$$

How OFDM Works: Frequency

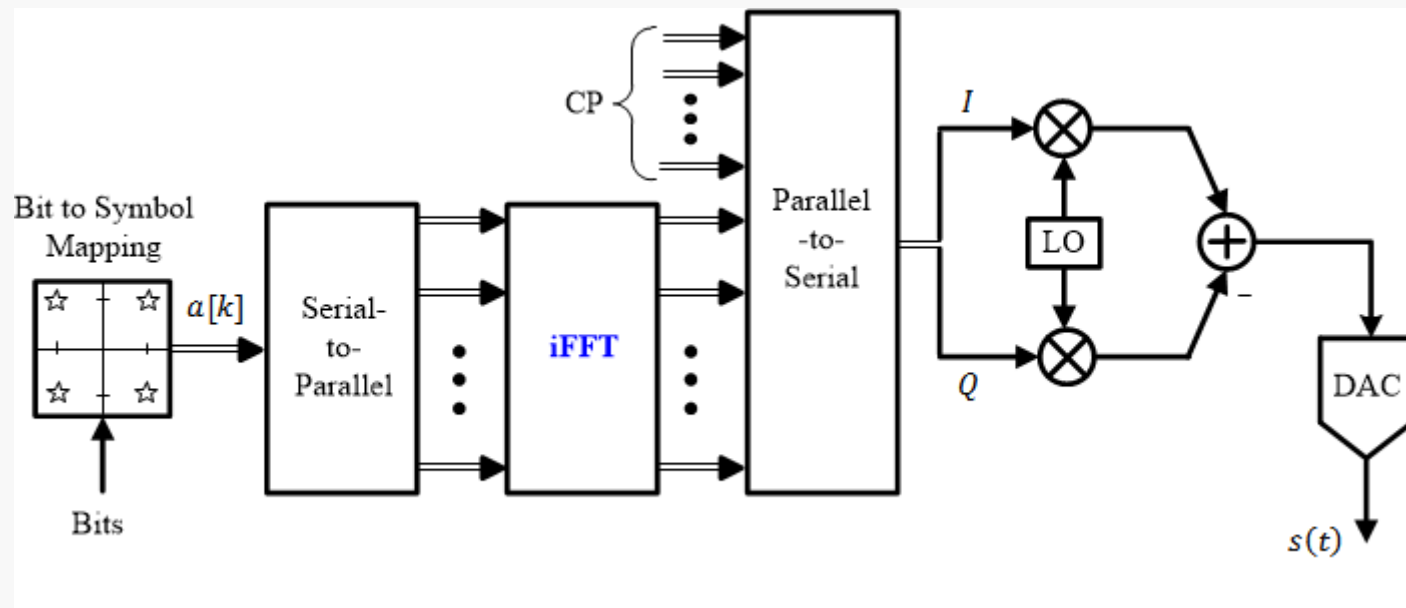
Invention of sliced bread radically changed the processing in 1928

- OFDM also slices the spectrum for easier handling

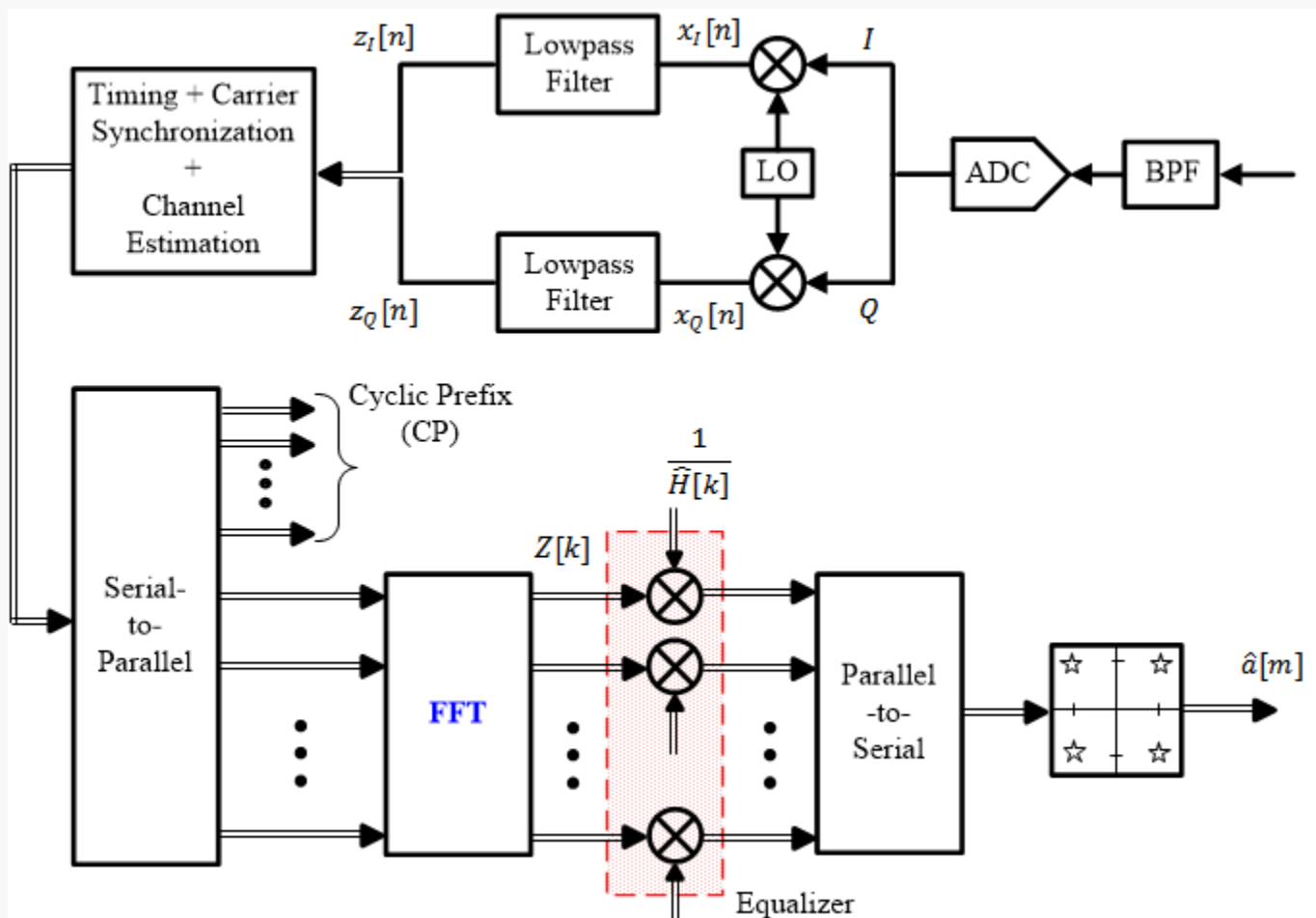


Summary: OFDM segments the available bandwidth of a frequency selective channel into several parallel frequency flat channels

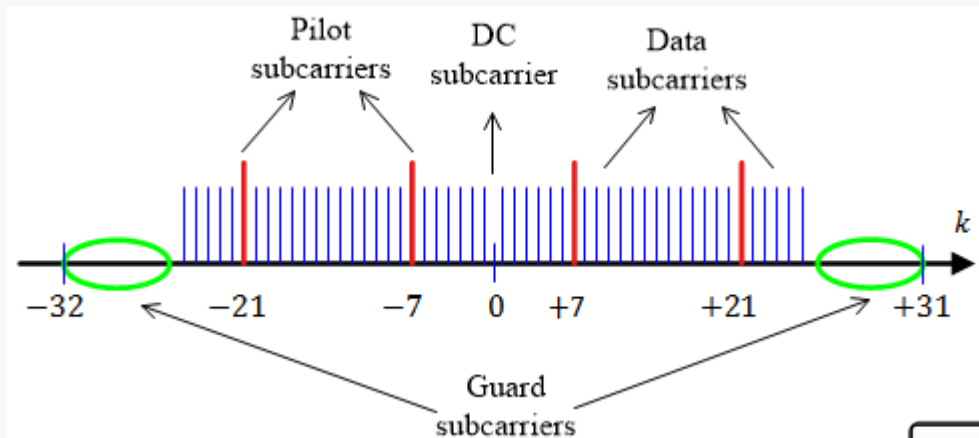
OFDM Tx



OFDM Rx

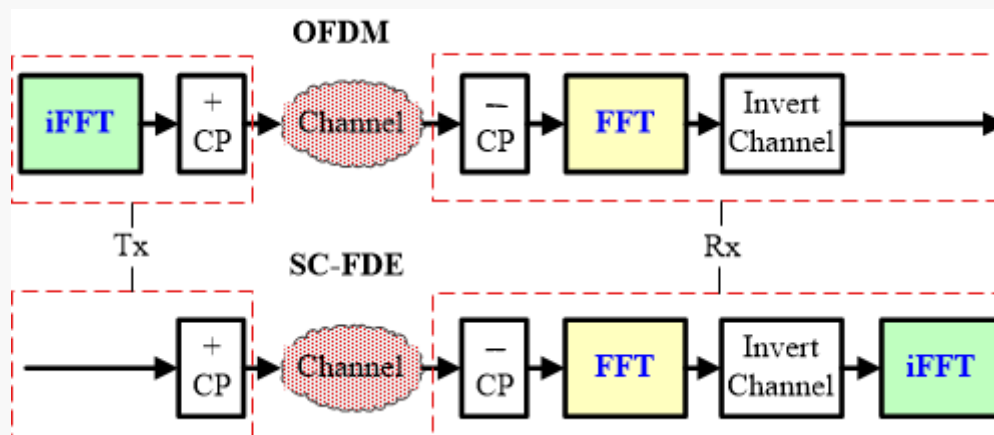


Example: IEEE 802.11a



Parameter	Value
Bandwidth $B = 1/T_S$	20 MHz
Data subcarriers	48
Pilot subcarriers	4
Null subcarriers	12
Total subcarriers	$48 + 4 + 12 = 64$
Subcarrier spacing $\Delta_F = B/N$	$20\text{MHz}/64 = 312.5 \text{ kHz}$
Useful symbol duration (FFT period) $T_u = 1/\Delta_F$	$3.2\mu\text{s}$
Cyclic prefix length T_{CP} (25% of T_u)	$T_u/4 = 0.8\mu\text{s}$
Total symbol duration T_{OFDM}	$3.2 + 0.8 = 4\mu\text{s}$

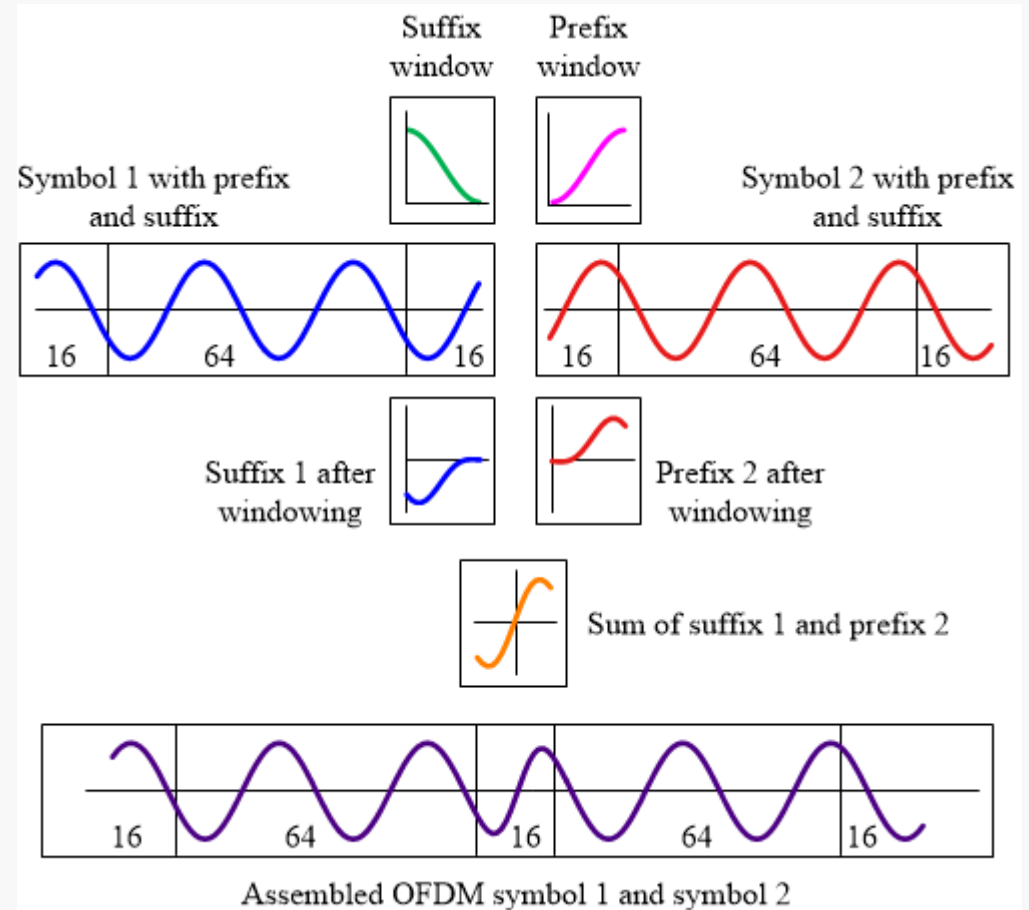
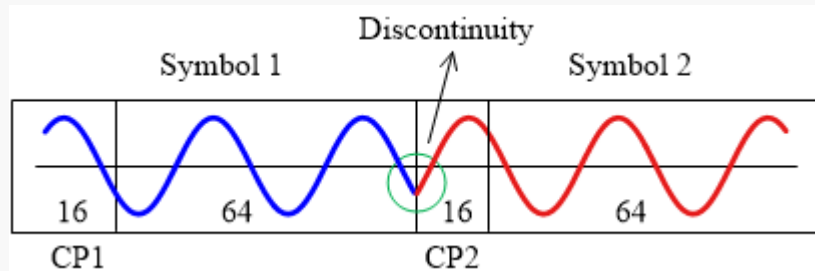
A Comparison with SC-FDE



Comparable performance in a coded system

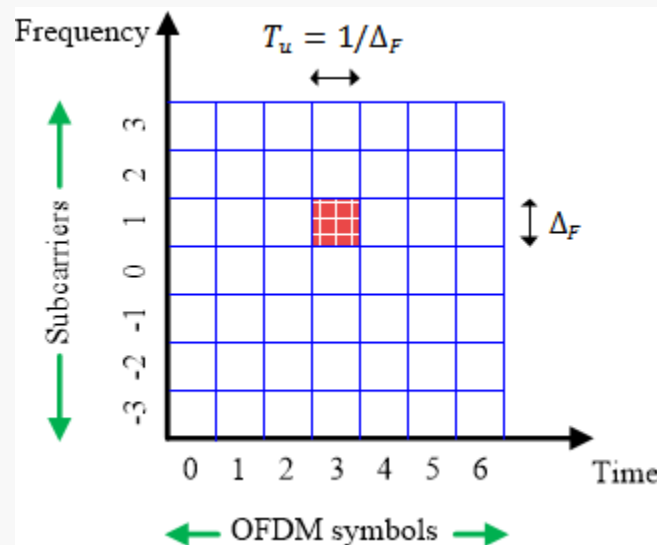
- OFDM is more sensitive to RF impairments (peak to average power ratio, carrier frequency offset, phase noise)
- Still the most widely used scheme, thanks to slicing the spectrum

Pulse Shaping in OFDM



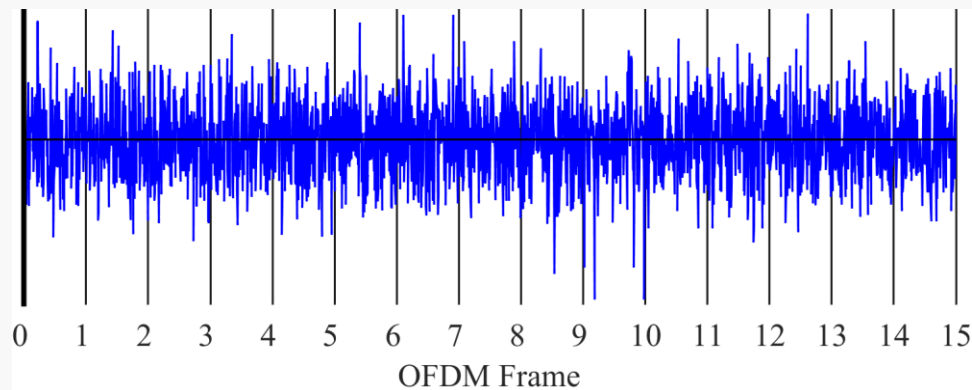
OFDM Time Frequency Grid

Thanks to orthogonality, slicing of the spectrum leads to handling of individual subcarriers



What Timing Synchronization Means

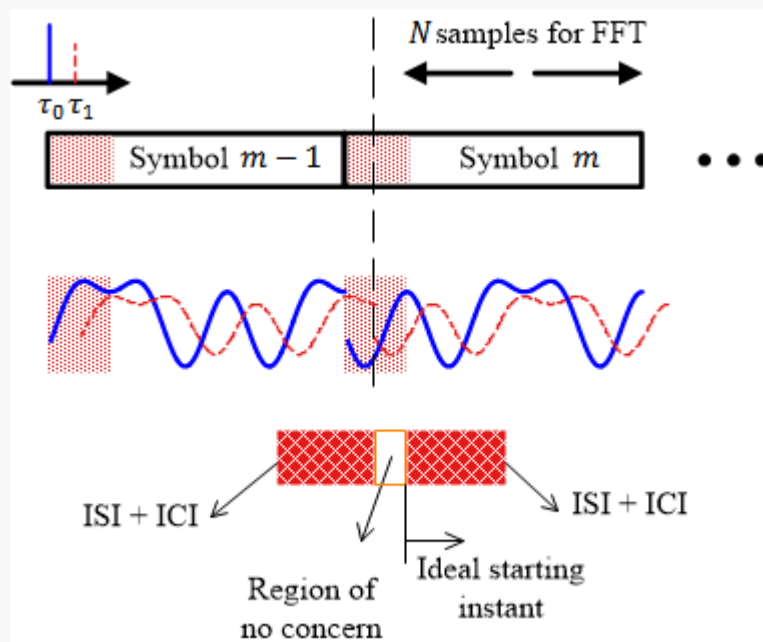
Marking the boundaries of OFDM symbols



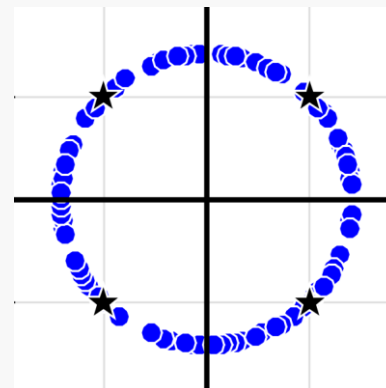
- **Integer STO:** starting sample for FFT input
- **Fractional STO:** not required!
 - Appears as additional phase rotation and hence becomes part of channel gain $H[k]$

Effect of Timing Mismatch

- There is an ISI-free region for timing estimation!



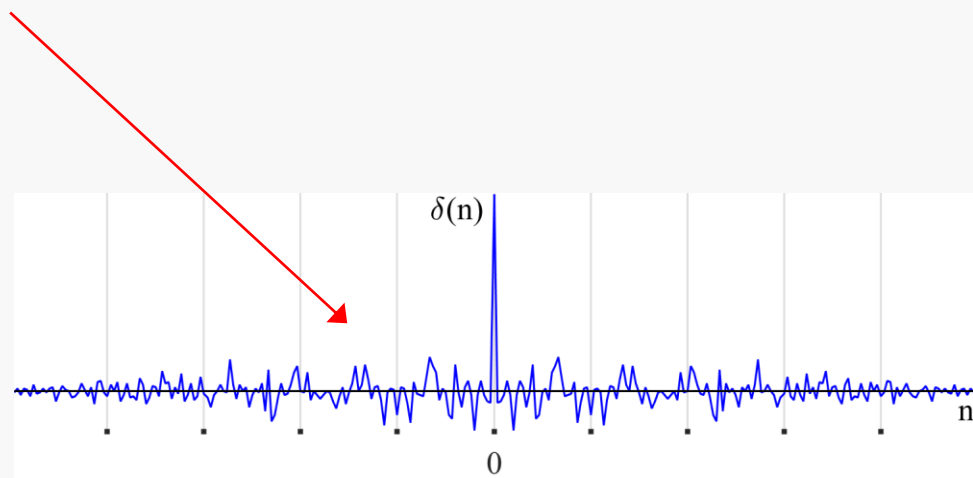
- From time-frequency duality, spinning of frequency domain symbols



Coarse Timing Synchronization

Autocorrelation of a Pseudo-Noise (PN) sequence resembles an impulse, just like that of white noise

- Zero for any time shift except when completely aligned

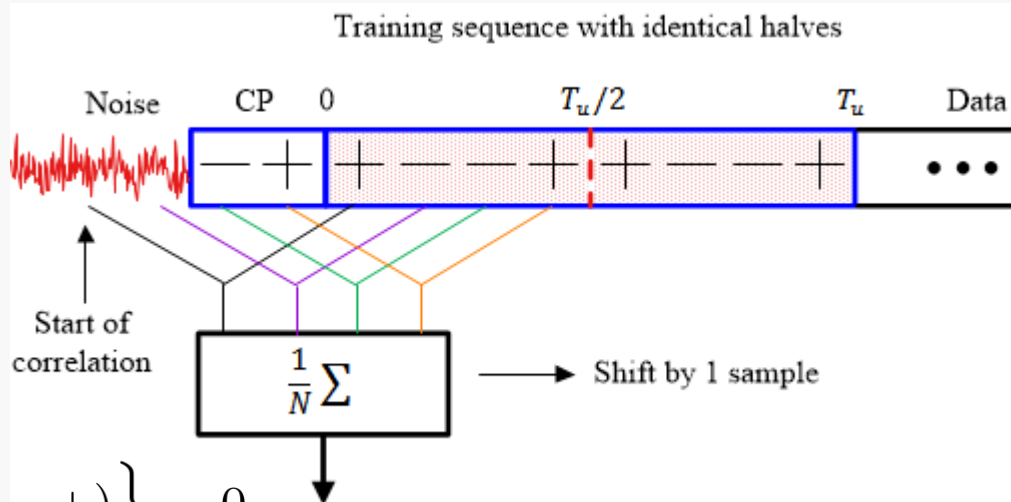


- Method: send a training sequence with two identical halves

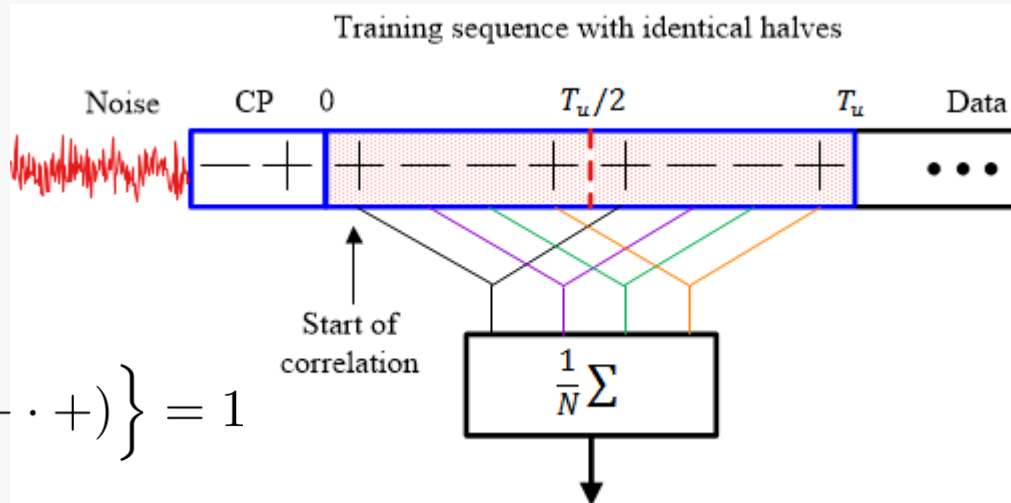
Coarse Timing Synchronization

Computing the autocorrelation

$$\frac{1}{4} \sum \{(- \cdot +), (+ \cdot -), (- \cdot -), (+ \cdot +)\} = 0$$



$$\frac{1}{4} \sum \{(+ \cdot +), (- \cdot -), (- \cdot -), (+ \cdot +)\} = 1$$



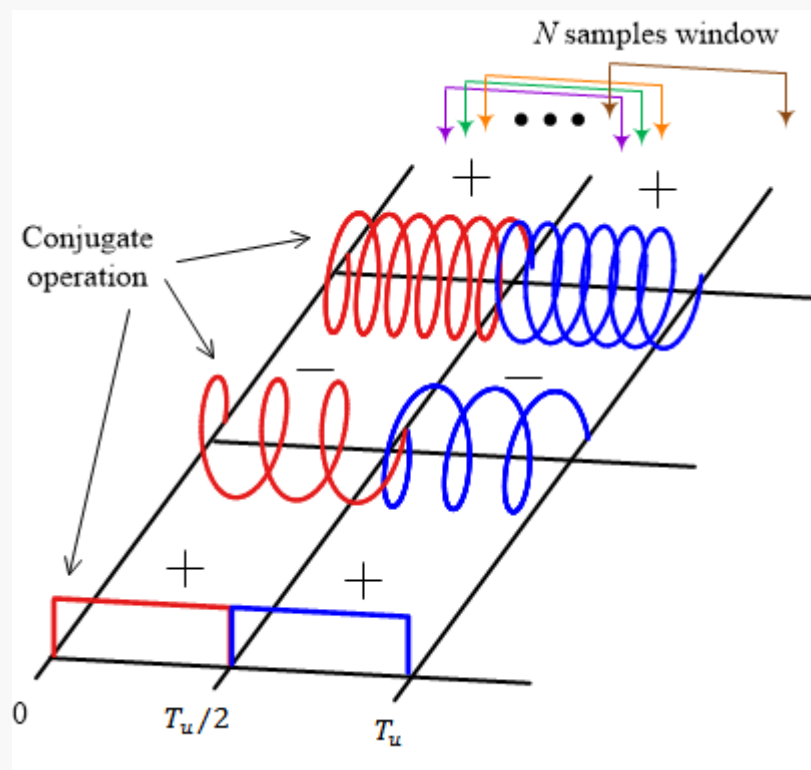
Coarse Timing Synchronization

Construction of a coarse timing metric

$$P(\varepsilon_0) = \sum_{i=0}^{N/2-1} z \left[\left(\varepsilon_0 + \frac{N}{2} \right) + i \right] z^* [\varepsilon_0 + i]$$

- Phase/frequency rotation
- Signal energy
- Timing metric

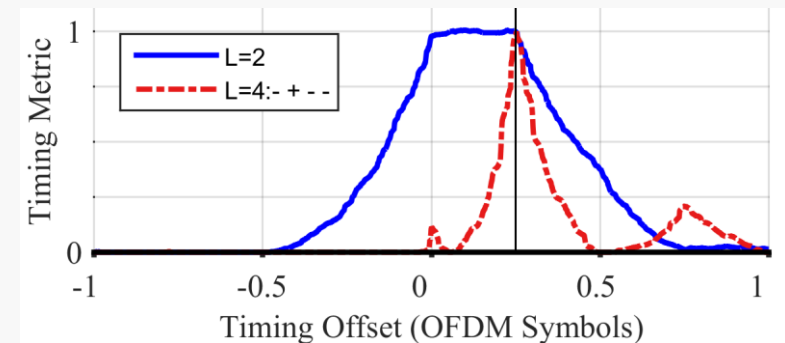
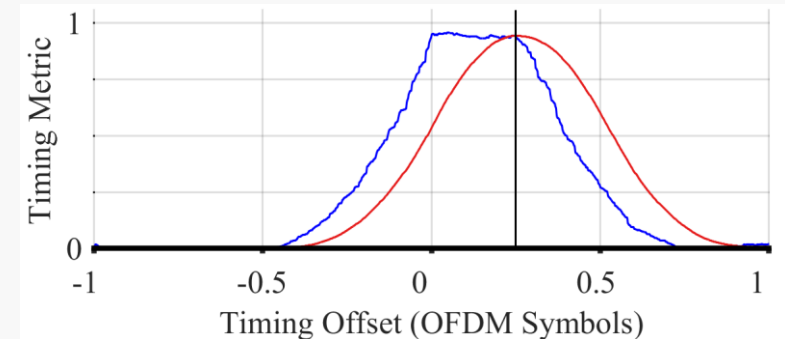
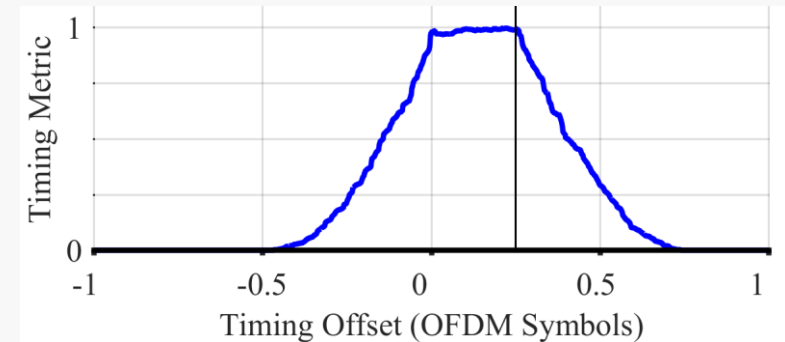
$$M(\varepsilon_0) = \frac{|P(\varepsilon_0)|^2}{[E(\varepsilon_0)]^2}$$



Coarse Timing Synchronization

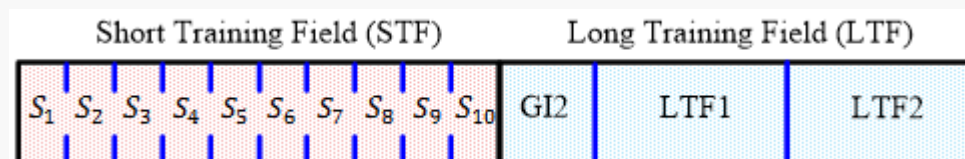
Improvements on the basic scheme

- Peak detection through 90% points
- Smoothing with a moving average filter
- Repetitions with different signs



Example: IEEE 802.11a

- **Short training field:** 10 repetitions used for signal detect, Automatic Gain Control (AGC), diversity selection, coarse timing and frequency synchronization
- **Long training field:** 2 repetitions used for fine frequency synchronization and channel estimation



Fine Timing Synchronization

Both identical halves are noisy

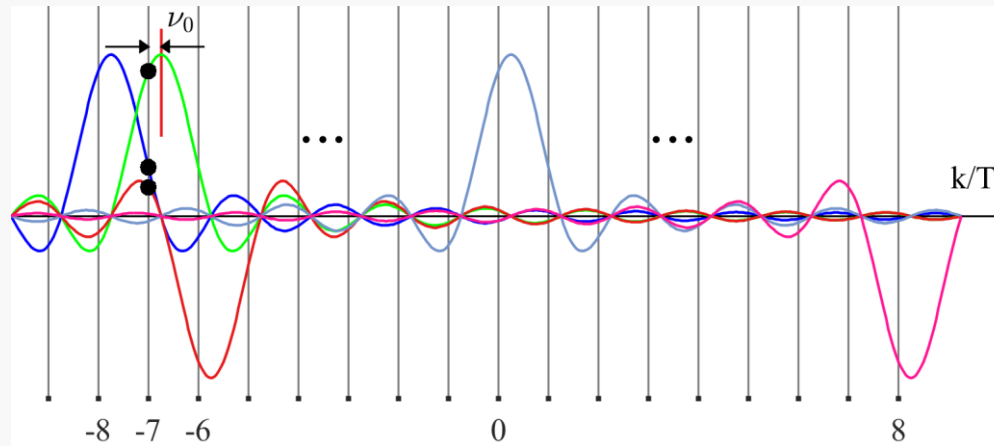
- Estimate and compensate for the coarse frequency offset
- Cross-correlation with a clean copy of the training sequence at the Rx
- Also known as matched filtering

$$\tilde{M}(\varepsilon_0) = \sum_{n=0}^{N-1} \tilde{z}[n + \varepsilon_0] s^*[n]$$

In continuous systems, this can also be done through averaging over several OFDM symbols

What Frequency Synchronization Means

- **Integer CFO:** FFT output will be shifted and symbol numbers will be wrong
- **Fractional CFO:** Inter-Carrier Interference (ICI) in a manner similar to ISI



Coarse Frequency Synchronization

Main idea: Frequency of any complex sinusoid can be estimated by taking a phase difference between any of its two samples at known time instants

$$\Delta\phi = 2\pi F(T_2 - T_1)$$

T_1 and T_2 are known while $\Delta\phi$ is estimated

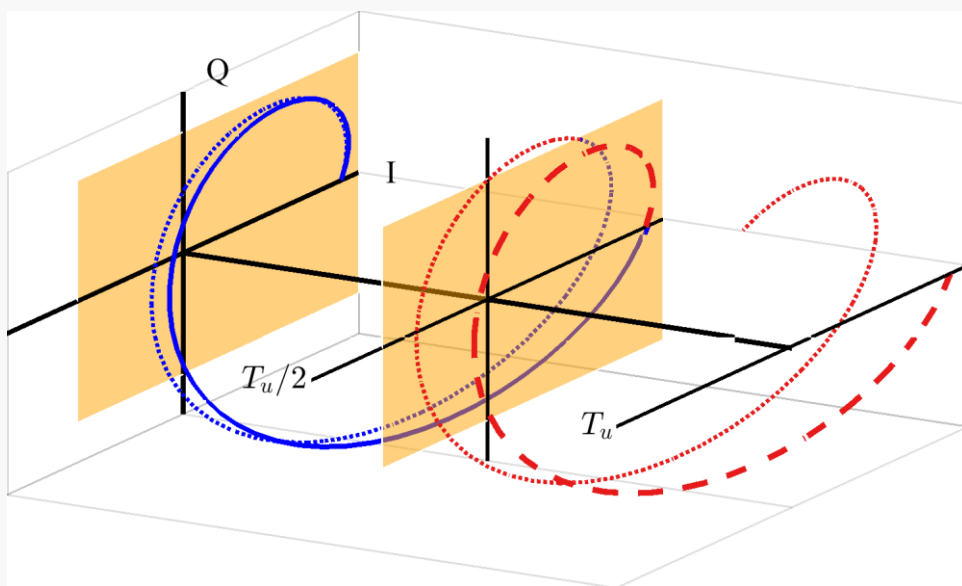
$$F = \frac{\Delta\phi}{2\pi(T_2 - T_1)}$$

In OFDM, we have not one but N such complex sinusoids!

Coarse Frequency Synchronization

- Recall $P(\varepsilon_0)$ from timing synchronization

$$P(\varepsilon_0) = \sum_{i=0}^{N/2-1} z \left[\left(\varepsilon_0 + \frac{N}{2} \right) + i \right] z^* [\varepsilon_0 + i]$$



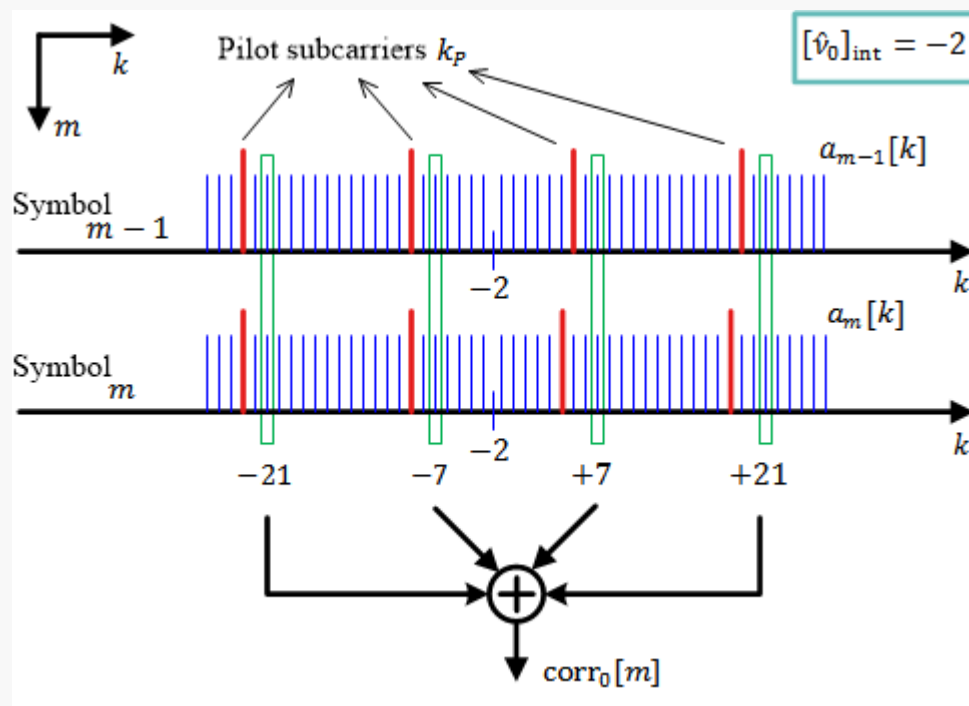
$$\hat{\nu}_0 = \frac{1}{\pi} \angle P(\hat{\varepsilon}_0)$$

Fine Frequency Synchronization

- Any of the techniques from single-carrier systems
- Autocorrelation with Long Training Sequence (LTF)

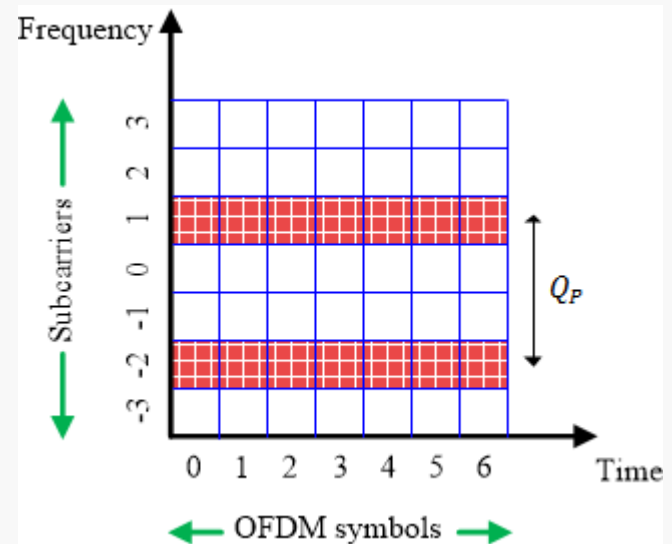
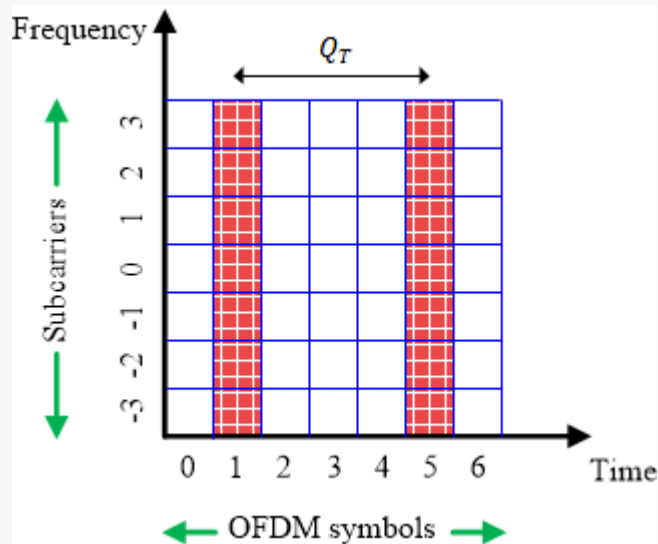
Integer CFO correction is done through correlation between adjacent OFDM symbols

$$\text{corr}_0[m] = \sum_{k_P} Z_m[k_P] \cdot Z_{m-1}^*[k_P]$$



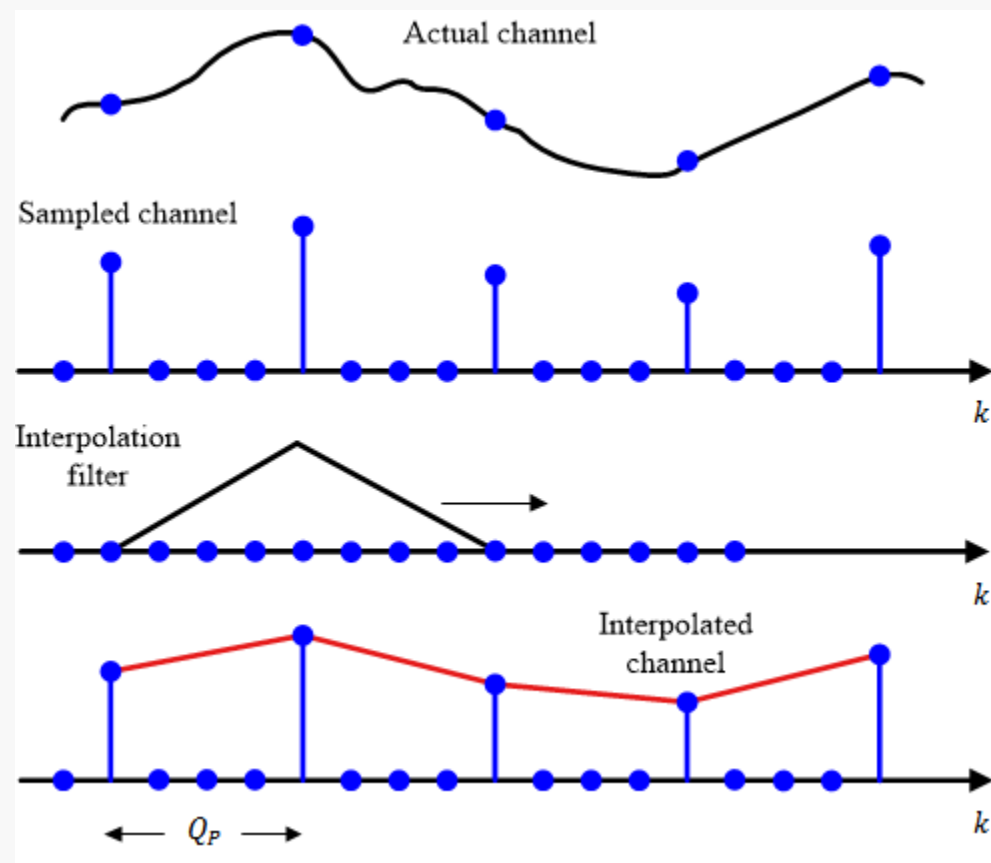
Channel Estimation

- Training based
- Pilots based



Channel Estimation

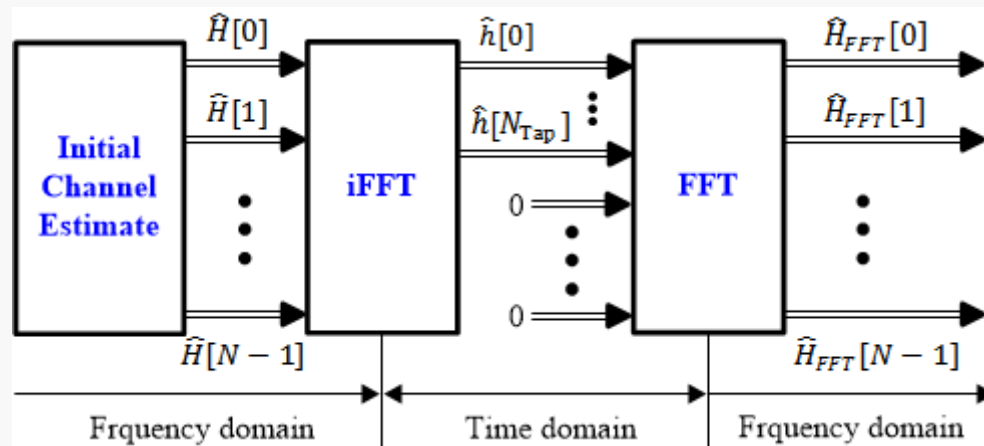
Linear interpolation between pilot subcarriers



Channel Estimation

- Transform domain technique

$$\hat{h}_{FFT}[n] = \begin{cases} h[n] + \text{noise} & n = 0, 1, \dots, N_{\text{Tap}} \\ 0 & n = N_{\text{Tap}} + 1, \dots, N - 1 \end{cases}$$



Channel Estimation

- Decision-directed estimation when channel is varying slowly

$$\hat{a}_m[k] = \frac{Z_m[k]}{\hat{H}_{m-1}[k]}$$

$$\hat{H}_m[k] = \frac{Z_m[k]}{\hat{a}_m[k]}$$

- Channel estimate expiry
- Error propagation